Weak 1 :- Introduction (Unit-1)

Introduction, introduction to python 1 & 2, introduction to Networkx-1 and Networkx 2, Social Network, The Challenge, Google page Rank, searching in a Network, link prediction, the contagions, importance of Acquaintace, Marketing on social N/w

Weak-2 :- Handling Real-world Network Datasets (Unit-1)

Introduction to Datasets, Ingredients Networks, Synonymy Network, Web Graph, Social Network Datasets, Datasets: Different formats, How to Download, Datasets Analysing Using Networkx, Analysing Using Gephi, Introduction : Emergence of connectedness, Advanced Material : Emergence of Connectedness, programming Illustration : Emergence of connectedness, summary to Datasets

Weak-3 :- Strength of Weak Ties (Unit-2)

Introduction, Granovetter's strenght of weak ties, Triads, clustering coefficient and neighborhood overlap, structure of weak ties, bridges and local bridges, validation of Granovetter's experiment using cell phone data, Embededness, structural Holes, social capital, finding communities in a graph (Brute Force Method), community Detection Using Girvan Newman Algorithm, visualising communities using Gephi, Tie strength, Social Media and passive Engagement, Betweenness Measures and Graph Partitioning, strong and weak Relationship -

Weak-4 :- Strong and Weak Relationships (continued) and Homophily Unit-2

Introduction to Homophily - should you watch your company?, selection and social Influence, Interplay betn Selection and Social Influence, Homophily - Definition and measurement, Foci Closure and Membership closure, Introduction to fatman Evolutionary model, fatman Evolutionary Model — The Base code (Adding people) fatman Evolutionary Model - The Base code (Adding social foci)

fatman Evolutionary Model - Implementing Homophily.

Quantifying the Effect of Triadic Closure.

fatman Evolutionary Model - implementing closures

fatman Evolutionary Model - implementing social Influence

fatman Evolutionary Model — Storing and analyzing longitudinal data

Weak - 5 :- Unit - 2 Homophily continued and ... of relationships

Spatial segregation : An Introduction
: Simulation of the Schelling Model
: Conclusion

Schelling Model Implementation : 1 (Introduction)
: 2 (Base code)
: 3 visualization and Getting a list of boundary & internal nodes)
: 4 (Getting a list of unsatisfied nodes)
: 5 (Shifting the unsatisfied nodes and visualizing the final graph.)

Positive & -ve Relationships : Introduction
Structural Balance,

Enemy's ENEMY IS A FRIEND, characterizing the structure of balanced n/w, Balance Theorem : proof of Balance Theorem, introduction to +ve & -ve edges, outline of implementation, creating graph, displaying it and counting unstable triangles, Moving a n/w from an unstable to stable state, forming two coalitions, forming two coalitions, visualizing coalitions & the evolution

Weak - 6 :- Unit - 3 (Link Analysis)

the Web Graph
Collecting the web graph
Equal Coin Distribution
Random coin dropping
Google page Ranking Using Web Graph
implementing page Rank Using points Distribution Method - 1
- 2
implementing page Rank using Random walk Method - 1
- 2

Degree Rank versus PageRank.

Week - 7 — Cascading Behavior in N/w (Unit - 3)
We follow, why do we follow?, Diffusion in N/w, modeling Diffusion, Impact of communities on Diffusion cascade and clusters, knowledge, thresholds & the collective Action, An Introduction to the programming screencast (coding 4 major idea). The Base code coding the first Big Idea - Increasing the payoff,
Second Big Idea - key people
third - Impact of communities on cascades
fourth - cascades & clusters

Weak 8 - (link Analysis) (continued) (unit 3)

Introduction to Hubs & Authorities (A Story)
Principle of Repeated Improvement (A Story)
Principle of Repeated Improvement (An example)
Hubs and Authorities
PageRank Revisited - An example
PageRank Revisited - convergence in the Example
- conservation and convergence
- matrix multiplication (Pre-requisite 1)
- Convergence in Repeated Matrix multiplication (Pre-requisite 1)
Addition of two vectors (pre-requisite 2)
Convergence in Repeated matrix multiplication - the Details
PageRank as a matrix Operation
Page Rank Explained

RRC
final topic

8000

Weak 9 — (Rich Get Richer phenomenon) (unit - 4)

Introduction to Powelaw
Why do Normal Distributions Appear?
Power law emerges in www graphs
Detecting presence of Powlaw
Rich Get Richer phenomenon
implementing Rich-getting-richer phenomenon (Barabasi-Albert Model) - 1 - 2

Weak 10 - (Power law (contd.) & Epidemics (unit - 4)

Rich getting richer phenomenon - 2
introduction to epidemics
simple Branching Process for modeling Epidemics
Basic Reproductive No
modeling epidemics on complex n/w
SIR and SIS spreading models
comparison betⁿ SIR & SIS spreading models
Basic Reproductive No Revised for complex N/W.
percolation model

Weak 11 Small world Phenomenon (unit 5)

small world Effect - An Introduction
milgram's Experiment

Weak 12 Pseudocore (How to go viral on web) (unit 5)

small world n/w
making homophily based edges